



1062CH12

AREAS RELATED TO CIRCLES

11

11.1 Areas of Sector and Segment of a Circle

You have already come across the terms *sector* and *segment* of a circle in your earlier classes. Recall that the portion (or part) of the circular region enclosed by two radii and the corresponding arc is called a *sector* of the circle and the portion (or part) of the circular region enclosed between a chord and the corresponding arc is called a *segment* of the circle.

Thus, in Fig. 11.1, shaded region OAPB is a *sector* of the circle with centre O. $\angle AOB$ is called the *angle* of the sector. Note that in this figure, unshaded region OAQB is also a sector of the circle. For obvious reasons, OAPB is called the *minor sector* and OAQB is called the *major sector*. You can also see that angle of the major sector is $360^\circ - \angle AOB$.

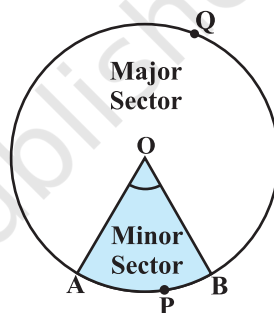


Fig. 11.1

Now, look at Fig. 11.2 in which AB is a chord of the circle with centre O. So, shaded region APB is a segment of the circle. You can also note that unshaded region AQB is another segment of the circle formed by the chord AB. For obvious reasons, APB is called the *minor segment* and AQB is called the *major segment*.

Remark : When we write ‘segment’ and ‘sector’ we will mean the ‘minor segment’ and the ‘minor sector’ respectively, unless stated otherwise.

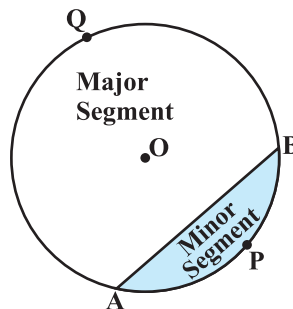


Fig. 11.2

Now with this knowledge, let us try to find some relations (or formulae) to calculate their areas.

Let OAPB be a sector of a circle with centre O and radius r (see Fig. 11.3). Let the degree measure of $\angle AOB$ be θ .

You know that area of a circle (in fact of a circular region or disc) is πr^2 .

In a way, we can consider this circular region to be a sector forming an angle of 360° (i.e., of degree measure 360) at the centre O. Now by applying the Unitary Method, we can arrive at the area of the sector OAPB as follows:

When degree measure of the angle at the centre is 360, area of the sector = πr^2

So, when the degree measure of the angle at the centre is 1, area of the sector = $\frac{\pi r^2}{360}$.

Therefore, when the degree measure of the angle at the centre is θ , area of the sector = $\frac{\pi r^2}{360} \times \theta = \frac{\theta}{360} \times \pi r^2$.

Thus, we obtain the following relation (or formula) for area of a sector of a circle:

$$\text{Area of the sector of angle } \theta = \frac{\theta}{360} \times \pi r^2,$$

where r is the radius of the circle and θ the angle of the sector in degrees.

Now, a natural question arises : Can we find the length of the arc APB corresponding to this sector? Yes. Again, by applying the Unitary Method and taking the whole length of the circle (of angle 360°) as $2\pi r$, we can obtain the required

length of the arc APB as $\frac{\theta}{360} \times 2\pi r$.

So, **length of an arc of a sector of angle θ** = $\frac{\theta}{360} \times 2\pi r$.

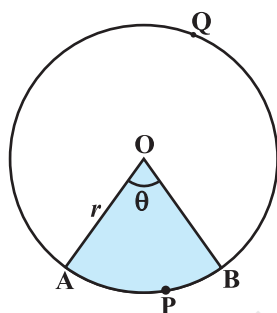


Fig. 11.3

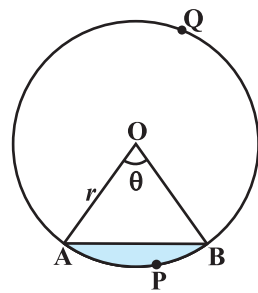


Fig. 11.4

11.2 Summary

In this chapter, you have studied the following points :

1. Length of an arc of a sector of a circle with radius r and angle with degree measure θ is

$$\frac{\theta}{360} \times 2\pi r.$$

2. Area of a sector of a circle with radius r and angle with degree measure θ is $\frac{\theta}{360} \times \pi r^2$.

3. Area of segment of a circle

= Area of the corresponding sector – Area of the corresponding triangle.